Technische Universität Braunschweig

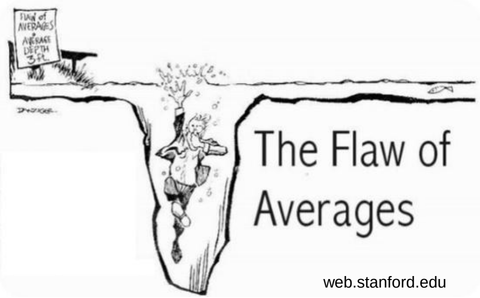
Decision Support
Institut für Wirtschaftsinformatik

Data Driven Decision Making

Lecture 4 – Dynamism

Recap and Motivation

- Uncertainty
- Information modeling
- Decision modeling
- Today: Can we react to observed changes in the information model?



Goals Today

- Dynamism in information and decision model
  - Dynamic vs. Static Planning
  - Dynamic decision process
- Stochastic and Dynamic Vehicle Routing
- First steps of modelling

Agenda

- Dynamism
- Case study: Urban Delivery and Traffic Management
- Preparation of Modelling

Dynamism

- Dynamic problems require not only one but several subsequent decisions.
- The decisions impact the future and are therefore interconnected.
- There is new information revealed after each decision was made.

- The solution for a dynamic problem is a strategy (policy)
- It defines how to react to newly revealed information.
- Reward or cost accumulate over time and decisions.

Static and Dynamic Planning

- Example: uncertain travel times

- Static planning:
  - Replanning not possible
  - Implementation of a (reliable) a-priori plan
  - Example: Bus

- Dynamic planning:
  - High impact of uncertainty requires replanning
  - Adaptations of plans
  - Example: Parcel delivery service

Examples for Dynamism

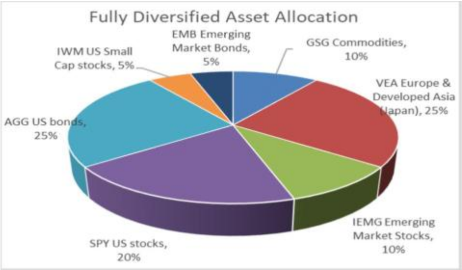
Urban Delivery

https://www.handelsblatt.com/unternehmen/handel-konsumgueter/onlinehandel-verkehrskollaps-per-mausklick/20604804.html?ticket=ST-190133-fsLx4SuFxff4v6EHdGk0-ap5

Line Production

https://www.venitec.de/de/leistungen-de/fertigungsplanung-fertigungsoptimierung/fliessfertigung

Asset Allocation



https://www.forbes.com/sites/randywarren/2020/08/06/maximizing-your-portfolio-benefits-via-asset-allocation/

Courier Service

https://www.qiez.de/top-5-kurierdienste-und-fahrradkuriere-in-berlin/

Inventory Management

https://www.lagertechnik-direkt.de/magazin/chaotische-lagerhaltung/

Games

https://www.forgerecycling.co.uk/blog/how-to-reuse-and-recycle-board-games/

Urban Mobility and Transportation:

Savelsbergh, van Woensel (Transportation Science, 2016):

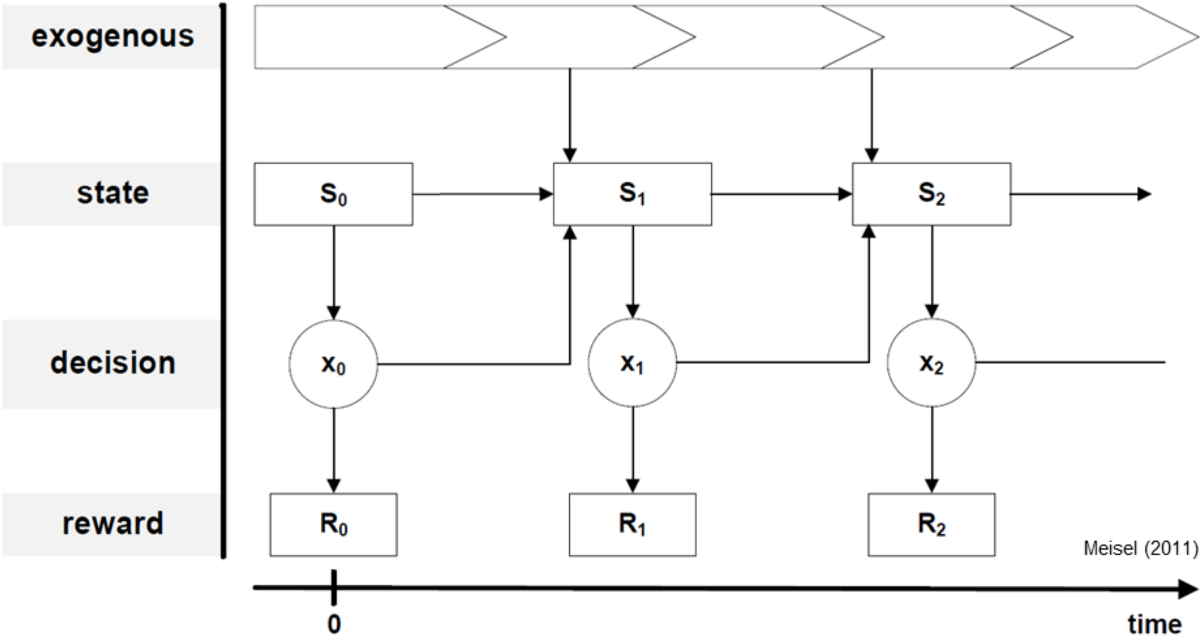
*“Highly **dynamic** and **volatile** decision making environments with access to massive amounts of near real-time data, which will be seen more and more in the future, may necessitate **new views, paradigms, and models for decision support**.”*

Speranza (European Journal of Operational Research, 2016):

*New challenge of “appropriately modeling **dynamic** events and simultaneously incorporating information about the **uncertainty** of future events.”*

→ Stochastic and Dynamic Vehicle Routing

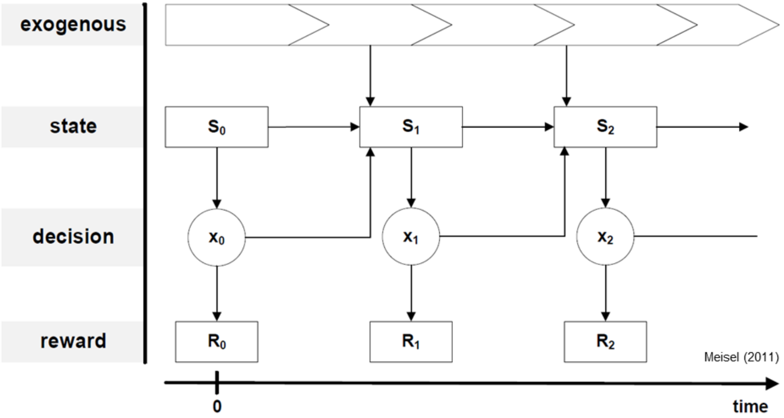
Components of a Dynamic Decision Problem



Meisel (2011)

Ingredients of a Dynamic Decision System

- Exogeneous: Development that cannot be controlled by decision maker (information model)
- Decision: Endogenous decision-making process, provides reward or costs (decision model)
- Both exogeneous and endogenous processes interact via decision states

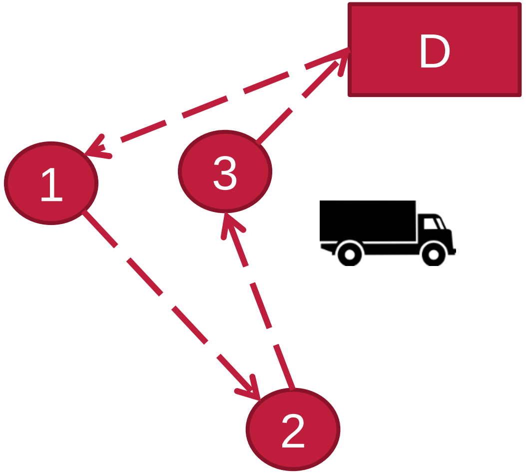


Meisel (2011)

Urban Delivery

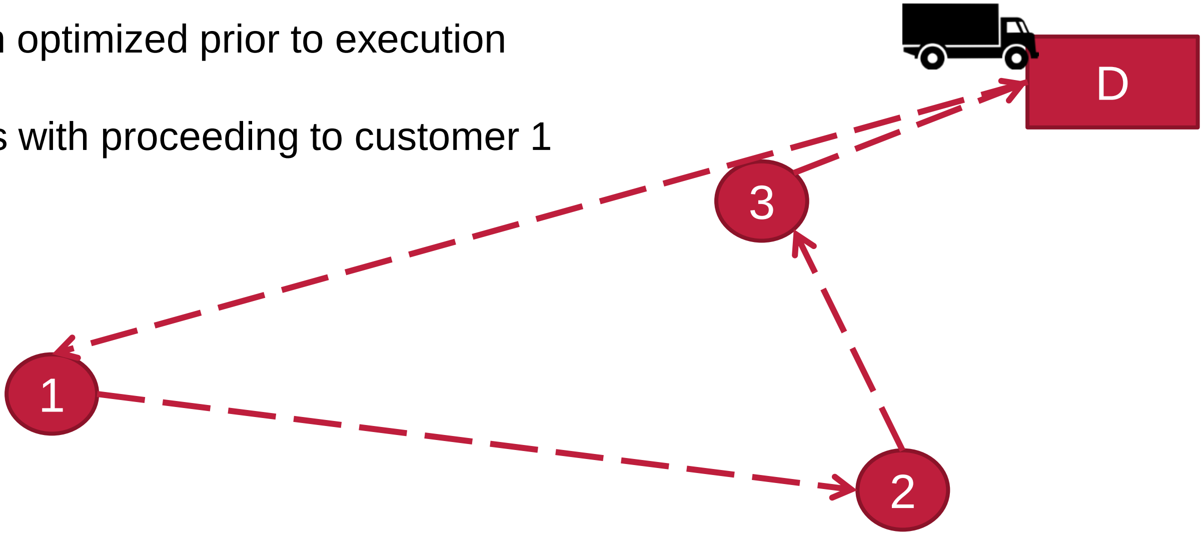
- Vehicle routing problem
- All customers must be served
- One depot, one vehicle
- Travel times vary over the day

- Exogenous: Varying travel times
- State: Customers to serve, Position of vehicle
- Decision: Whom to visit next
  or how to update the tour
- Reward: Total travel time



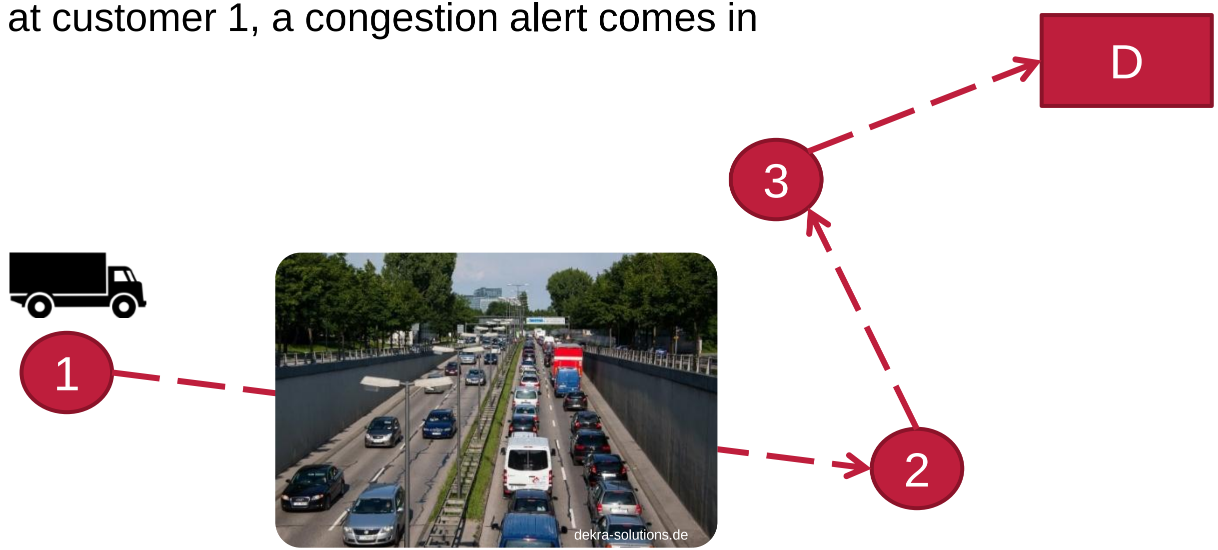
Example: Urban Delivery

- Route has been optimized prior to execution
- Execution starts with proceeding to customer 1



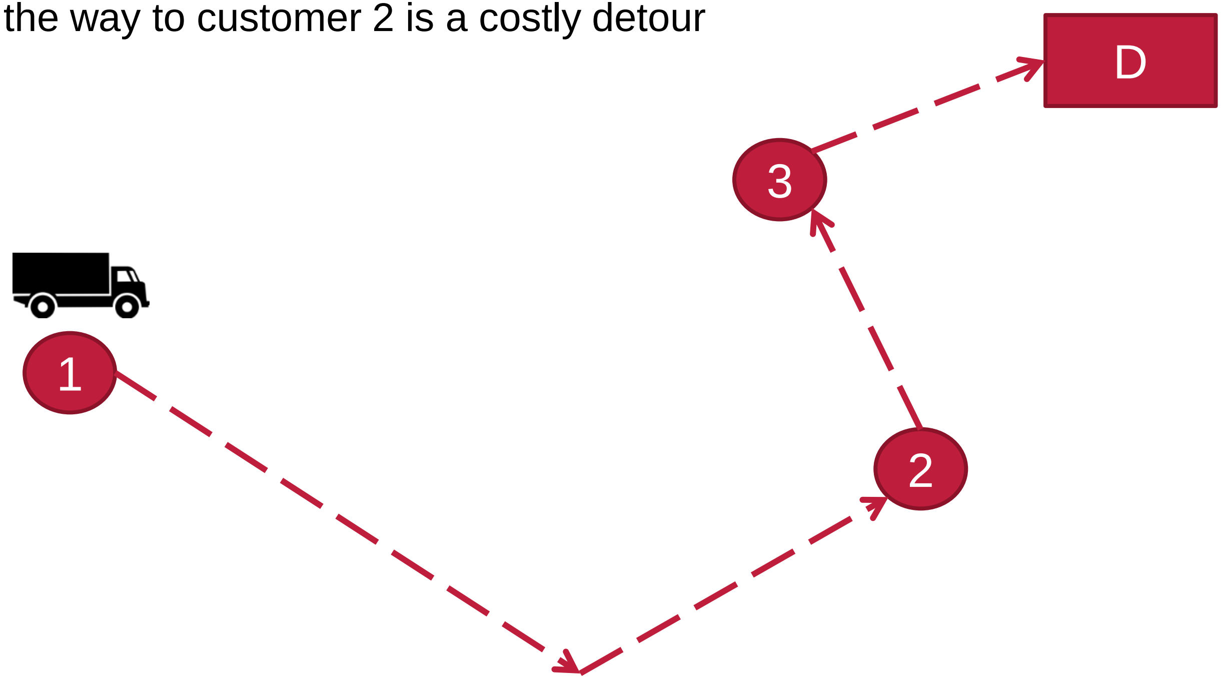
Example: Urban Delivery

- Having arrived at customer 1, a congestion alert comes in



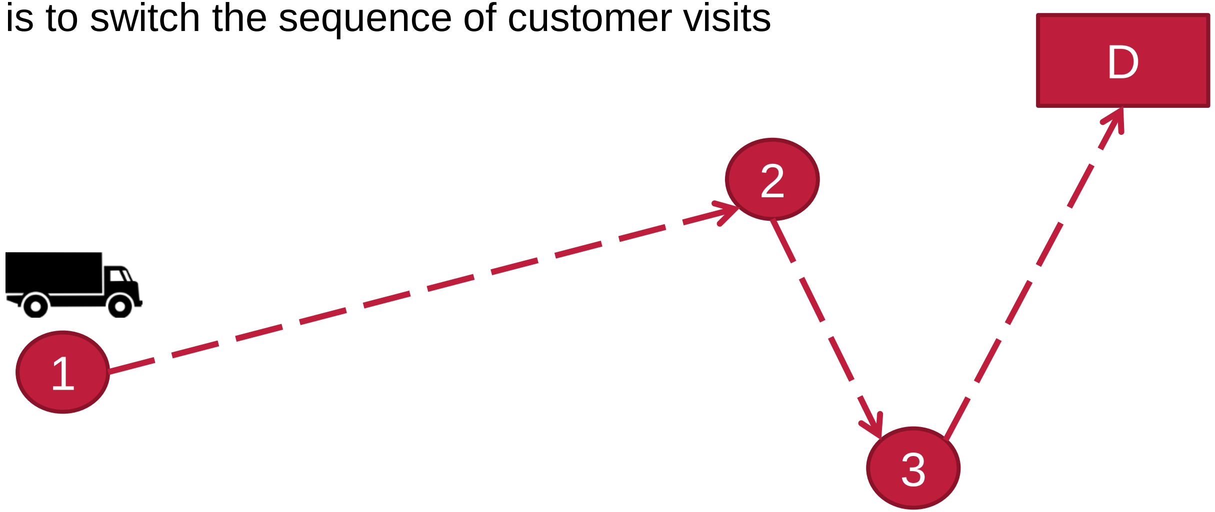
Example: Urban Delivery

- One option on the way to customer 2 is a costly detour

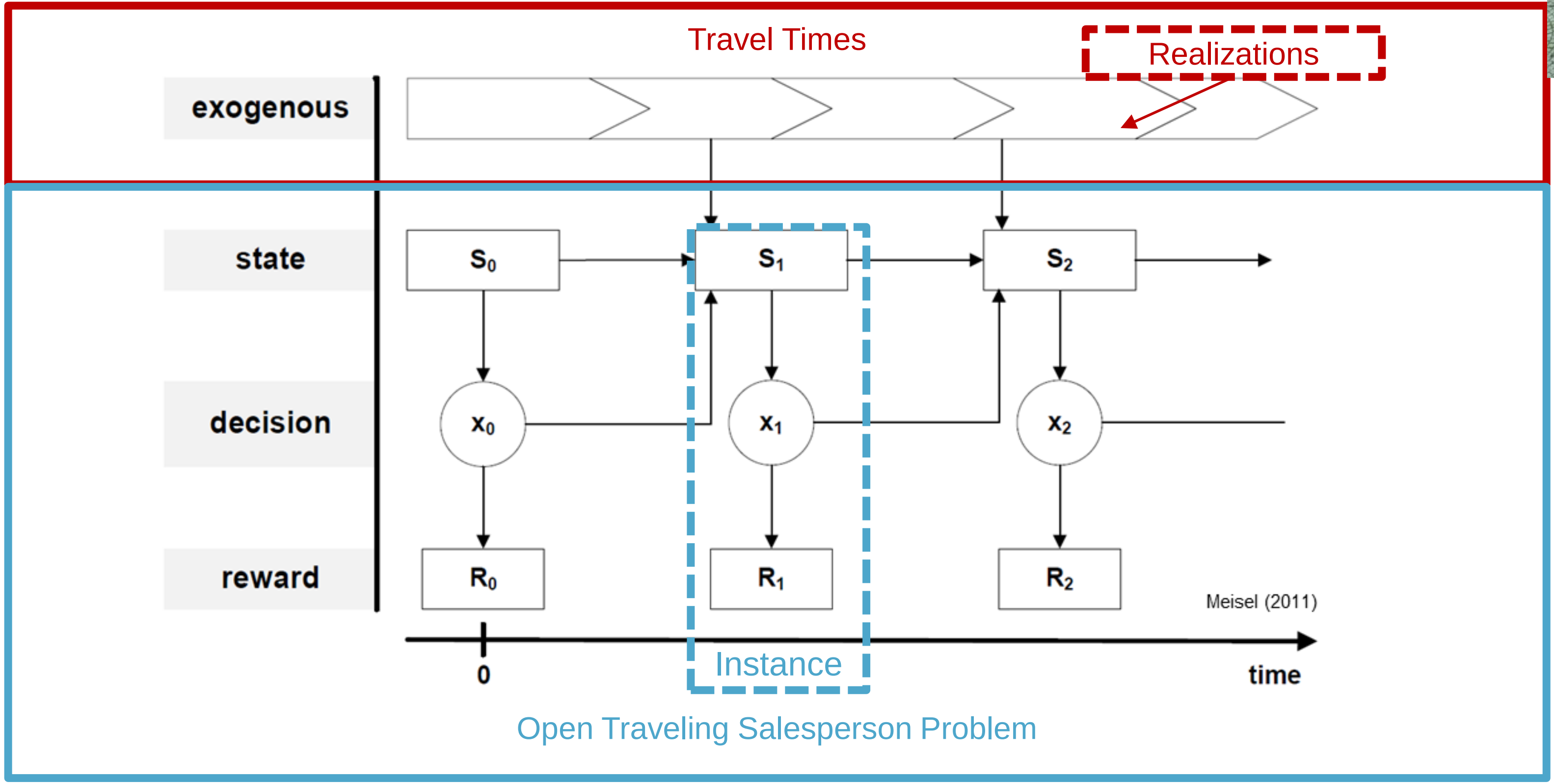


Example: Urban Delivery

- Another option is to switch the sequence of customer visits



Dynamic Decision Process



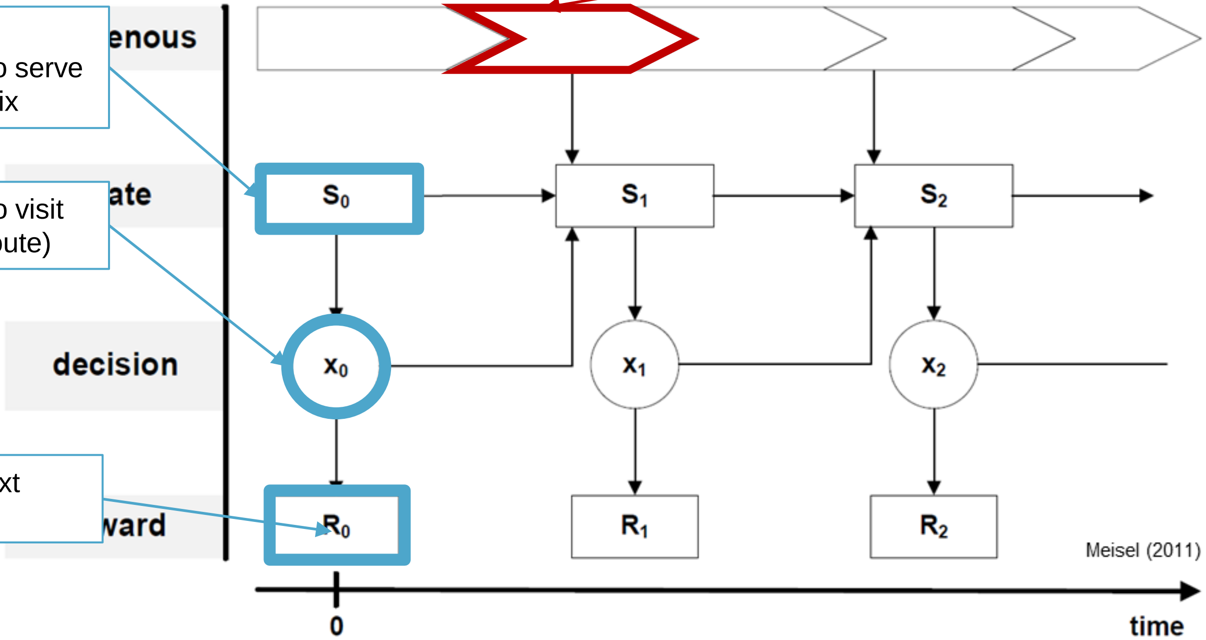
Dynamic Decision Process

Change in travel time matrix

- Position
- Customers left to serve
- Travel time matrix

- Next customer to visit (new tentative route)

- Travel time to next customer



Dynamic Decision Process

- Position
- Customers left to serve
- Travel time matrix

- Next customer to visit (new tentative route)

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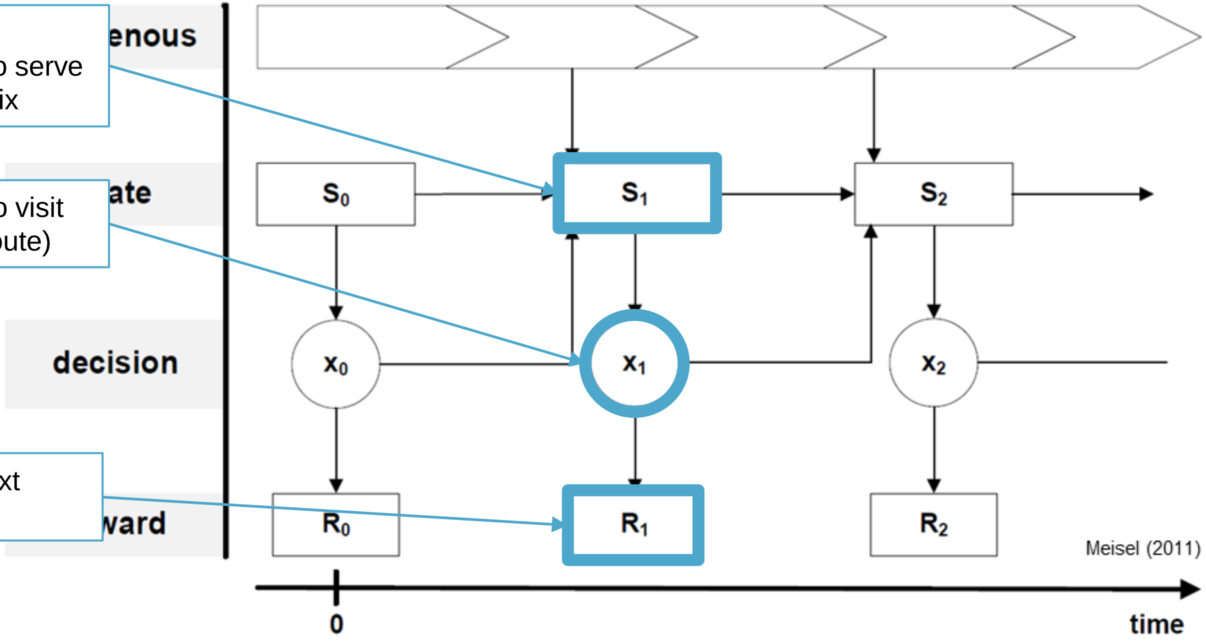
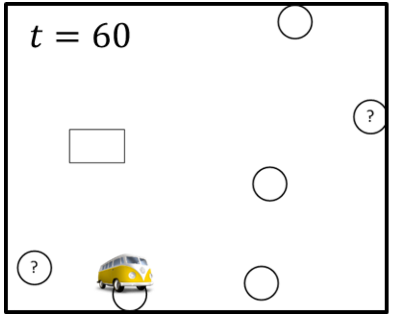


Courier Services

- Pickup and delivery problem
- Customer request service
- Requests are accepted/rejected
- Reward: Total no. of customer served

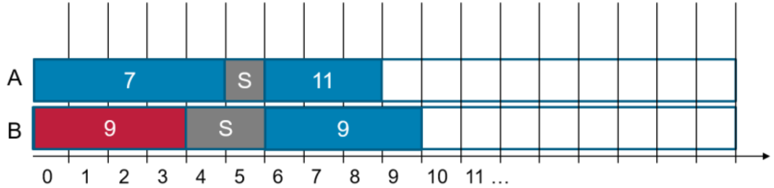
- Exogenous: Customer requests
- State: Customer accepted, position of vehicle
- Decision: Which request to accept, whom to visit next
- Reward: Newly accepted customers



Line Production

- Scheduling problem
- Problem horizon
- Set of machines
- Arriving jobs: duration, deadline, type
- Sequence dependent set-up times

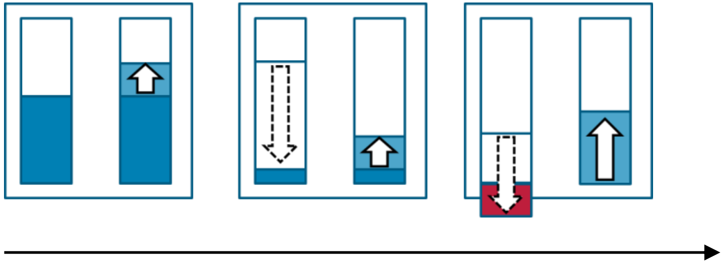
- Exogenous: Arriving jobs
- State: Current schedule, new jobs
- Decision: Sequence, machine
- Reward: Min. total delay



Inventory Management

- Inventory problem
- Horizon in terms of days
- Initial inventory and max capacity
- Uncertain demand per day
- Delivery cost
- Holding costs per unit and day
- If demand>inventory, difference is served by (expensive) third party

- Exogenous: uncertain demand
- State: current inventory
- Decision: order quantity
- Reward: min. total costs



Agenda

- Dynamism
- **Case study: Urban Delivery and Traffic Management**
- Preparation of Modelling

Case Study

- Urban Delivery

Felix Köster, Marlin W. Ulmer, Dirk C. Mattfeld, Geir Hasle,Anticipating emission-sensitive traffic management strategies for dynamic delivery routing, Transportation Research Part D: Transport and Environment, Volume 62, 2018, Pages 345-361

- Traffic Management

- Stochastic travel times

- Question: Can dynamic replanning save travel times?


News › World › Europe

Rome and Milan ban cars from roads in battle against air pollution

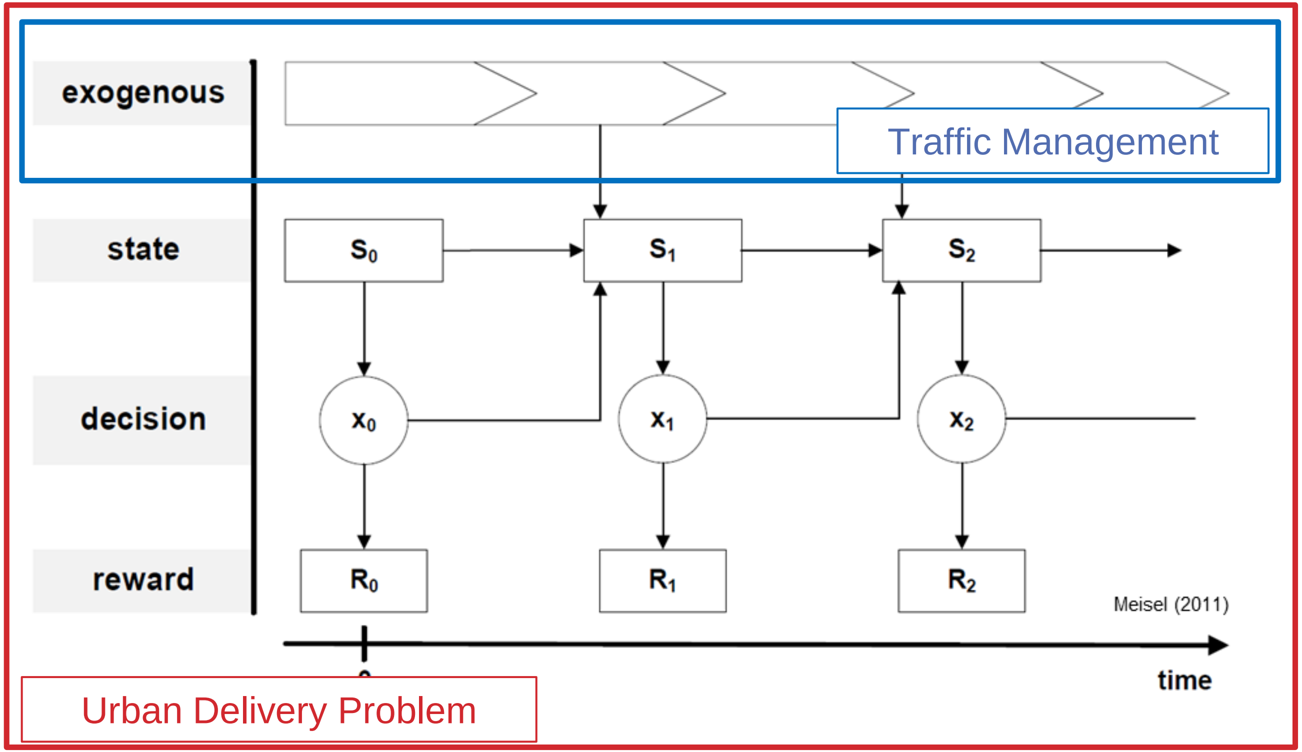
Smog levels in the Italian cities have exceeded healthy levels for more than 30 consecutive days

Siobhan Fenton | @siobhanfenton | Tuesday 29 December 2015 | 0 comments

75 shares

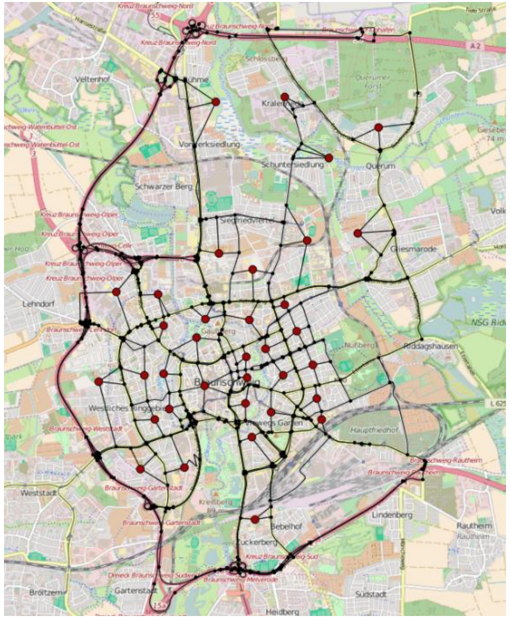
Smog levels in the Italian cities have exceeded healthy levels for more than 30 consecutive days AP

Case Study


exogenous
Traffic Management
state
S_0
S_1
S_2
decision
x_0
x_1
x_2
reward
R_0
R_1
R_2
Meisel (2011)
time
Urban Delivery Problem

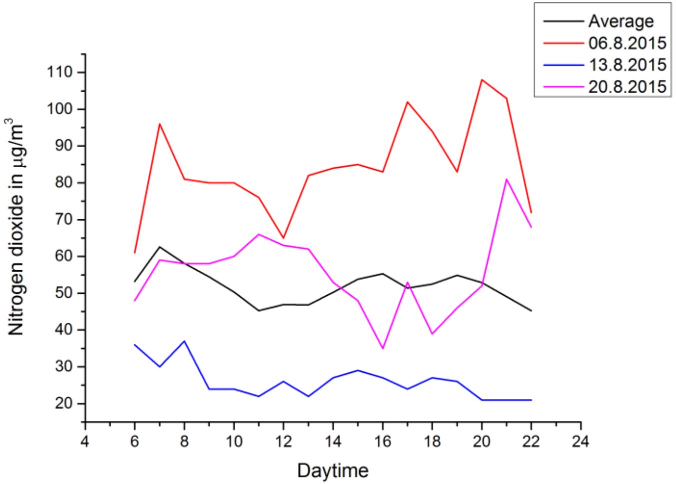
Motivation: Delivery service

- Parcel delivery
- Last-Mile Delivery
- 50% of total transport costs
- Vehicle fleet
- Vehicles deliver parcels in urban districts
- Initial distribution of parcels to vehicles
- Individual routing decision



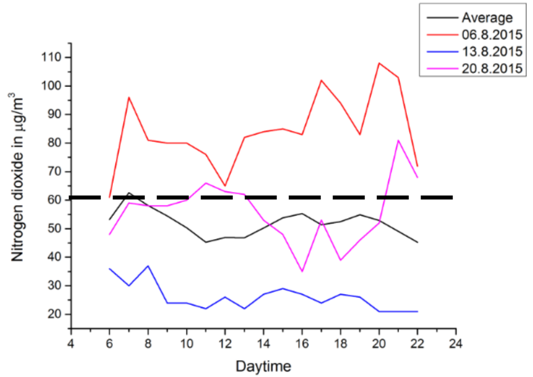
Motivation: Traffic Management

- Controls traffic flows within the city
- Goals:
  - Enable efficient mobility
  - Social and political guidelines
- Emissions:
  - Volume
  - Air pollution
- EU regulation on nitrogen oxides
- Stations:
  - at hotspots
  - measure nitrogen oxides



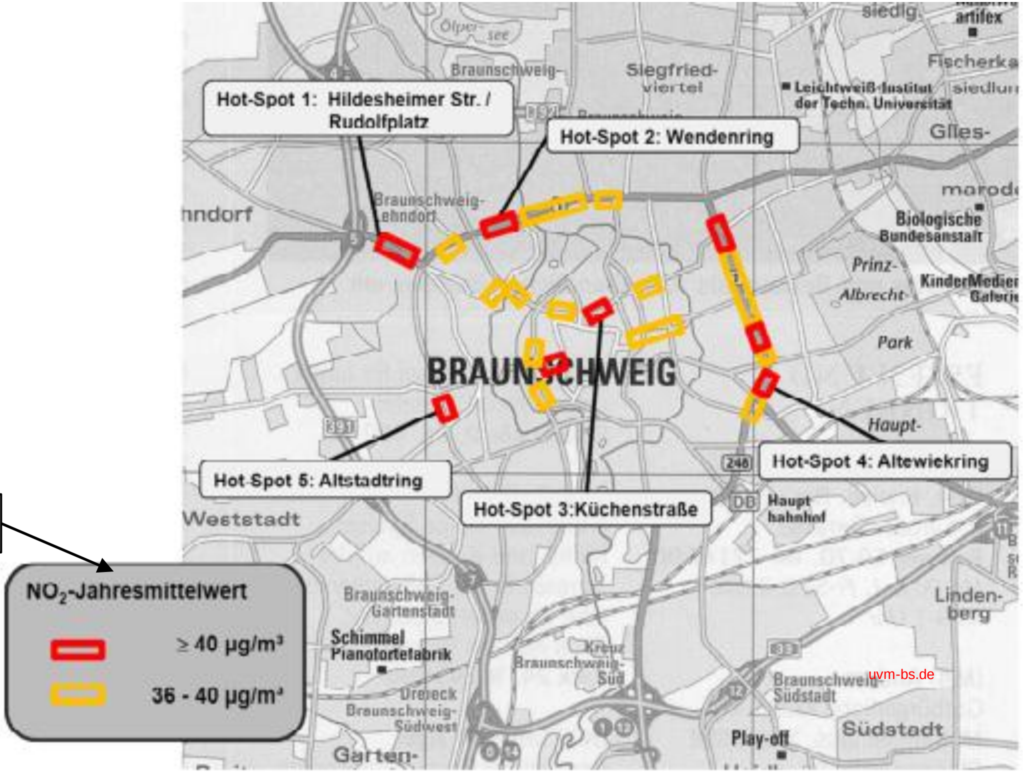
Dynamic Traffic Management

- Constant measurement of air pollution
- If NO2 pollution reaches 60g/m3:
  - Avoidance of further inflows
  - Enabling fast outward flows
- Tools:
  - Traffic lights
  - Changes in road capacity
  - Change in speed limits



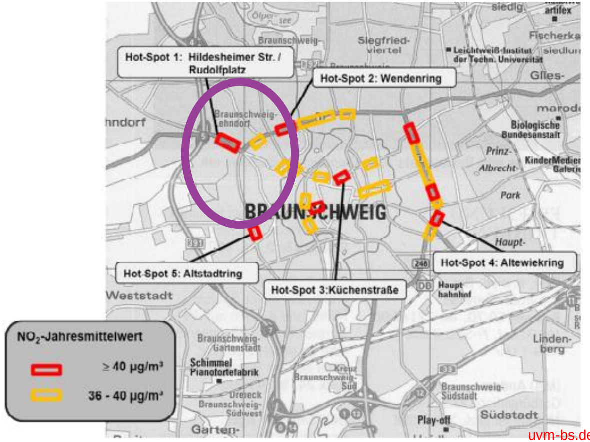
Traffic Management Braunschweig

Yearly average

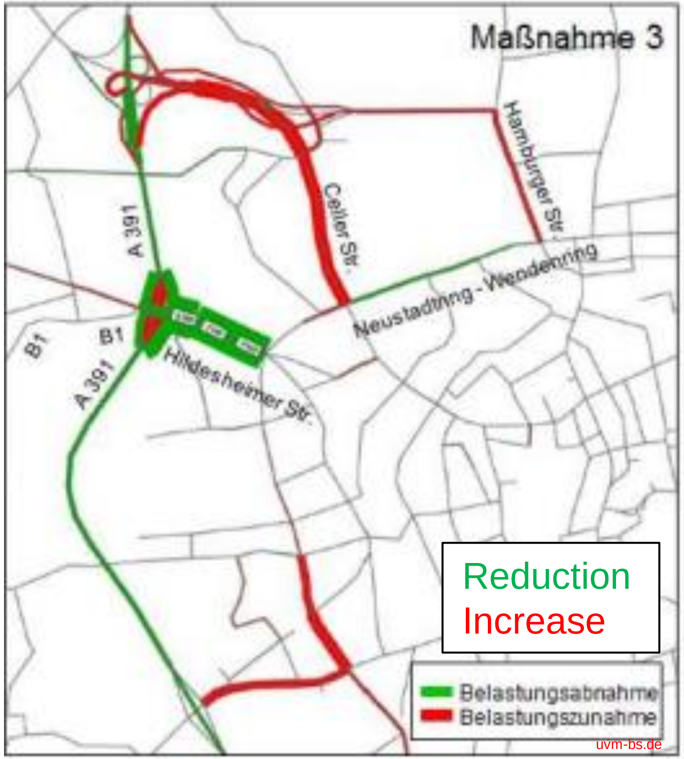


Hot-Spot 1: Hildesheimer Str.

- Limit traffic from A391
- Suggest exits at Celler, Münchner, and Hamburger Straße.



uvm-bs.de



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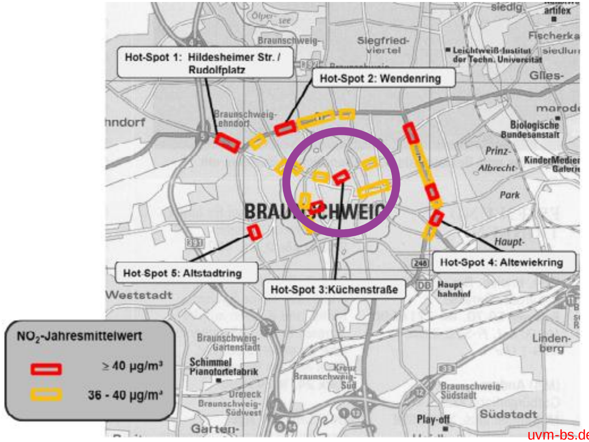
Hot-Spot: Downtown

- Closure Fallersleber Straße



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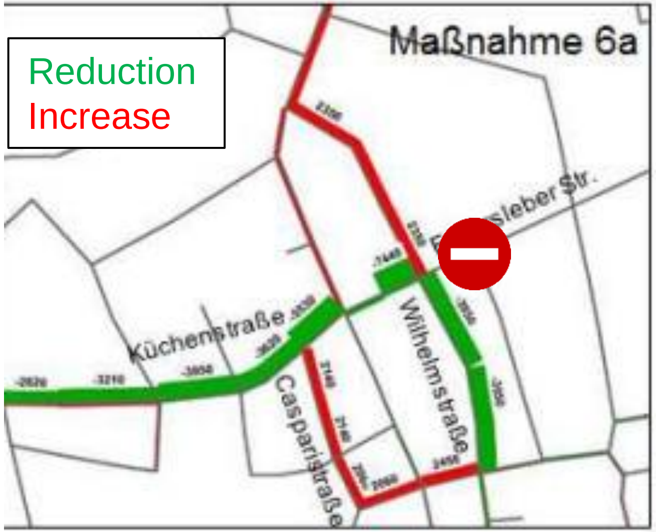
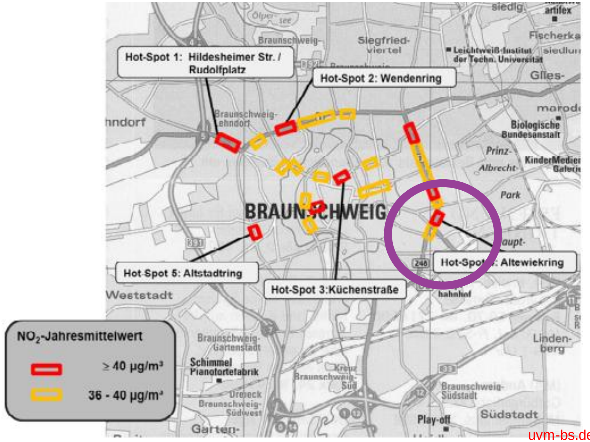


Hot-Spot 4: Altewiekring

- Limit left turn at Schillstraße



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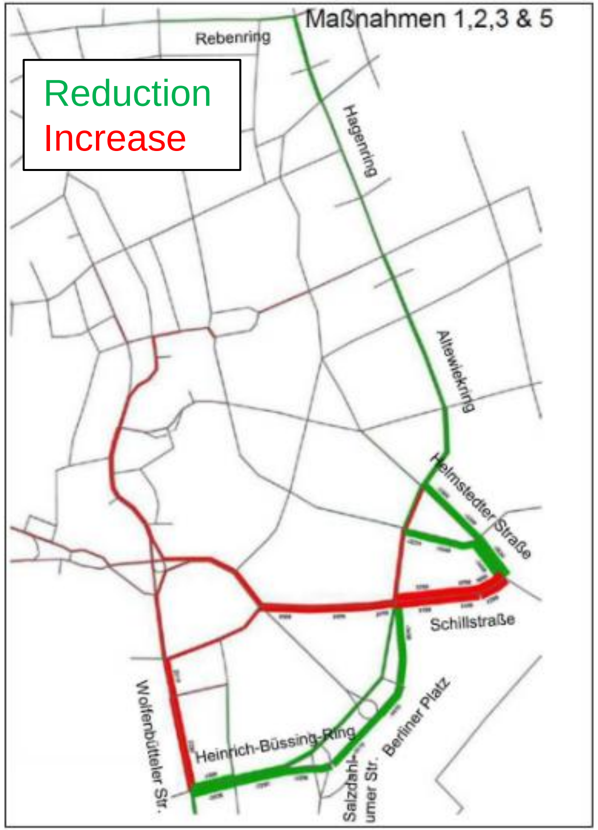
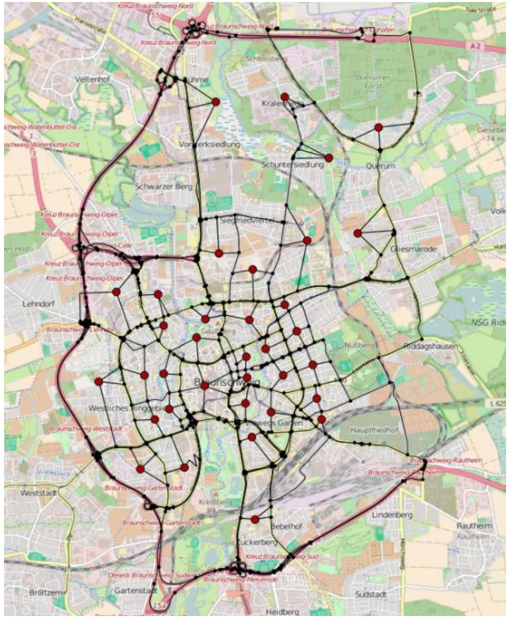


Delivery Service



- Vehicle fleet
- Vehicles deliver packages in urban districts
- Initial distribution to vehicles
- Individual routing decision
- Information Model: Stochastic travel times due to traffic management decisions

Traffic Management and Travel Times

Traffic Management controls street capacities

↓

Capacities impact traffic flows

↓

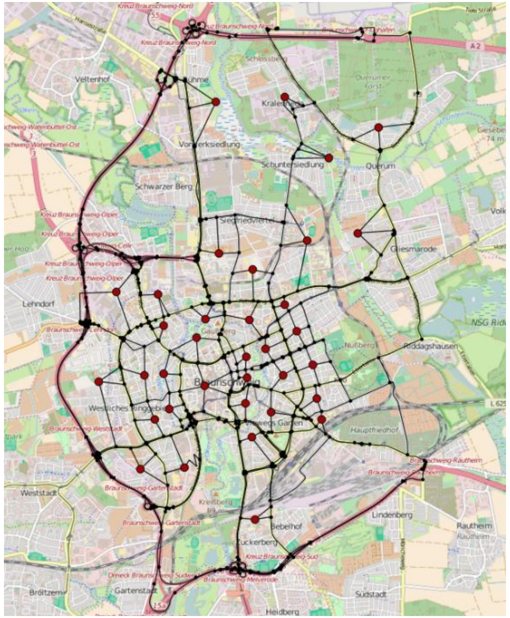
Traffic flows define speed per street segment

↓

Shortest paths and travel time matrices

Delivery service

- Vehicle fleet
- Vehicles deliver packages in urban districts
- Initial distribution to vehicles
- Individual routing decision
- Information Model: Stochastic travel times due to traffic management decisions
- Decision Model: Where to go next after serving a district (how to adapt the tentative route)

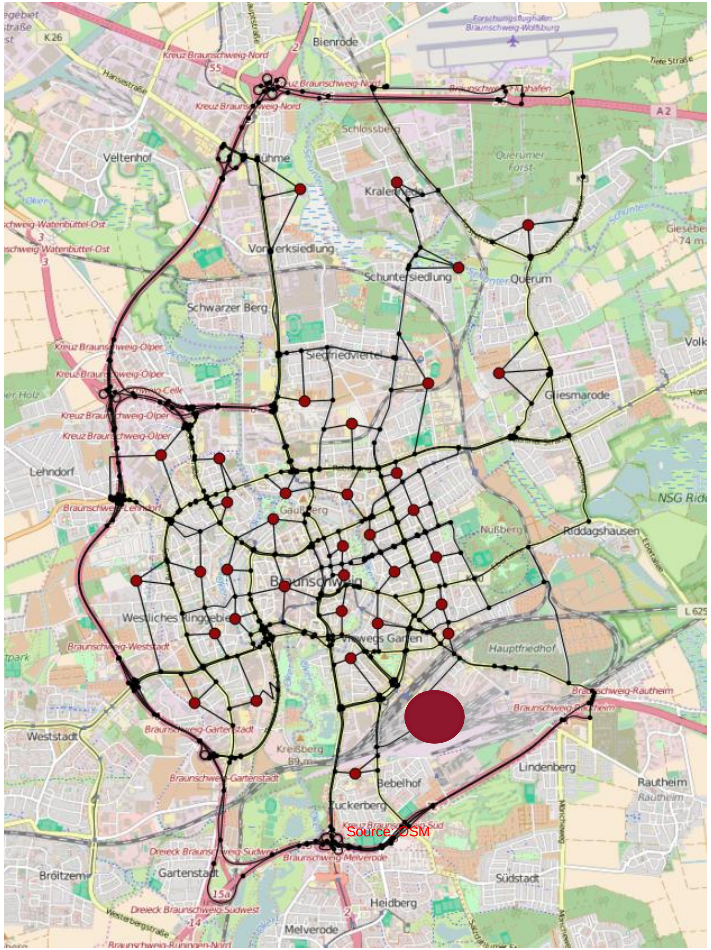


Case Study Overview

- Instance Generation
- Solution Method
- Results

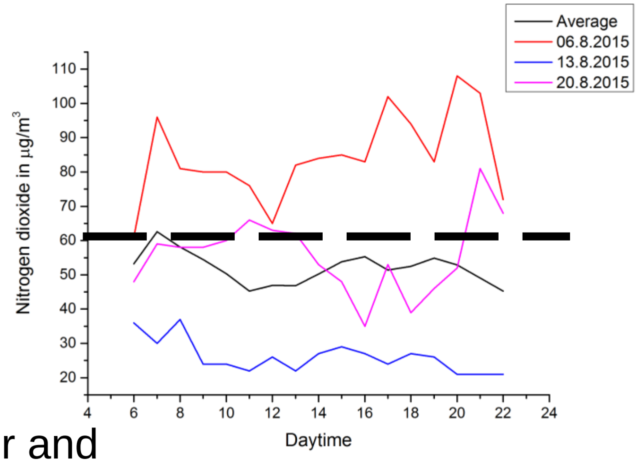
Instances

- 3 Hotspots in Braunschweig
- Change of hotspot transport strategy
  - Min. 1 hour active
  - Each hotspot is independent
  - 8 Traffic strategies
- Vehicle speed:
  - City roads: 25 km/h
  - Reduced: 10 km/h
  - Increased: 30 km/h
  - Autobahn: 80 km/h
- Depot:



Emission Data

- Data set of the State of Lower Saxony:
  - 3 months hourly values for NO_2
  - 92 days
  - Standard deviation 3-11 micrograms/m3
  - 24% of time over 60 micrograms/m3
- Hotspots: Emissions data from Braunschweig, Hannover and Osnabrück
- If the pollution at a hotspot is greater than 60 micrograms/m3, the respective measures are taken



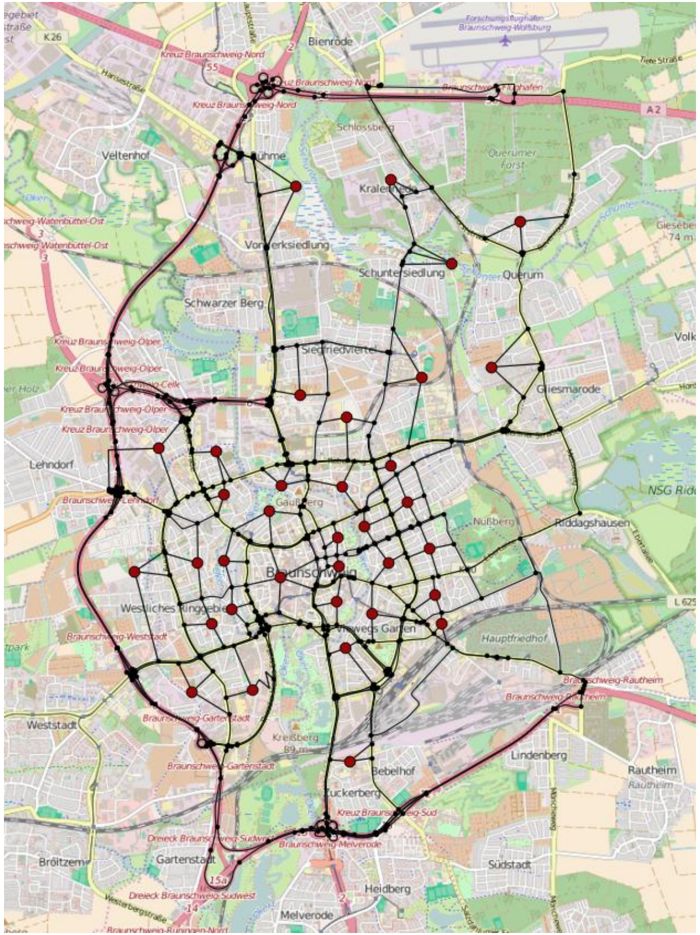
Case Study Overview

- Instance Generation
- **Solution Method**
- **Results**

Method

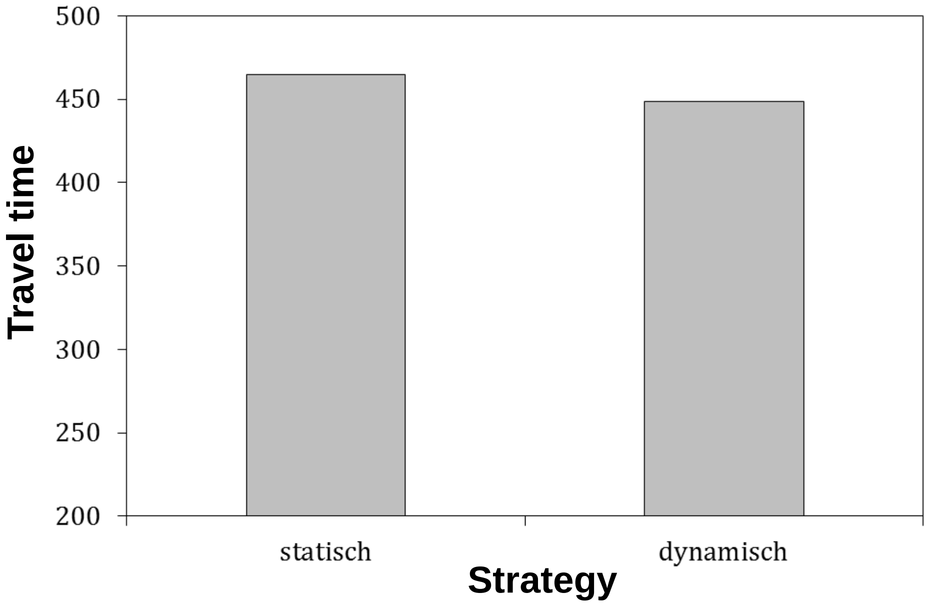
- Initial distribution: Vehicle Routing Problem with time limit

- For every vehicle in every decision state:
  - Calculate travel times: Dijkstra-Algorithm
  - Solve the open traveling salesperson problem

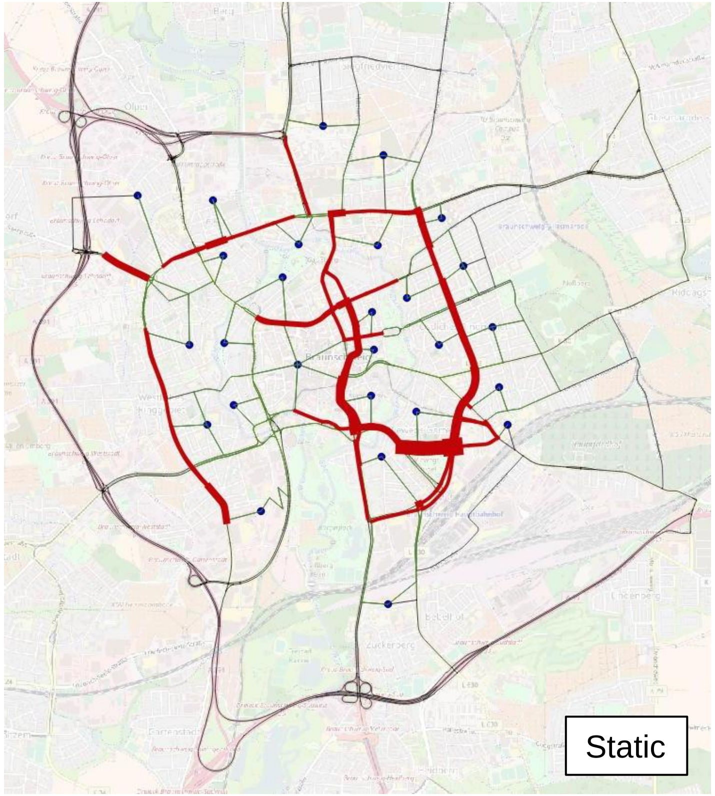


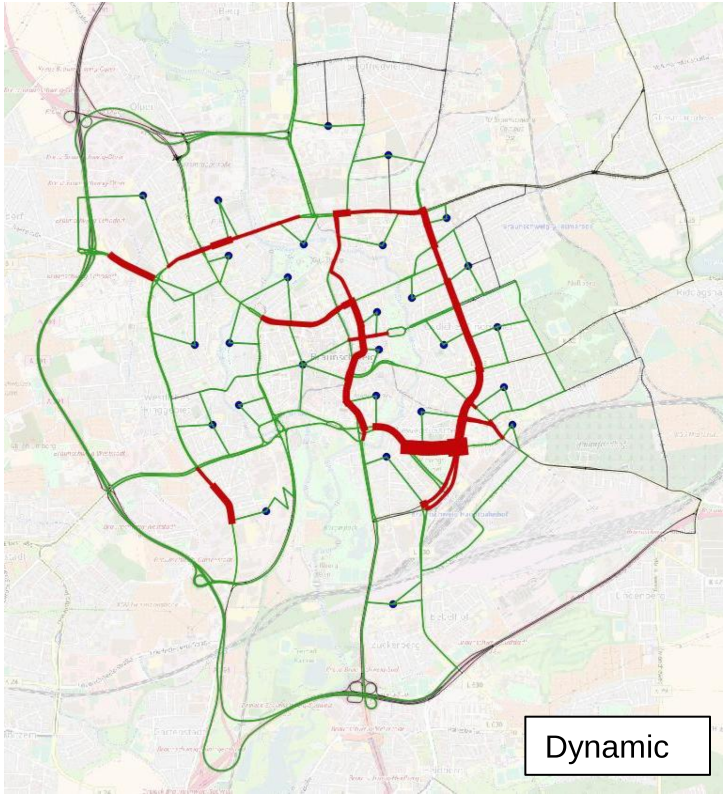
Results

- 1000 Days
- Average sum of travel times
- Reduction by 3.5%



Street Usage



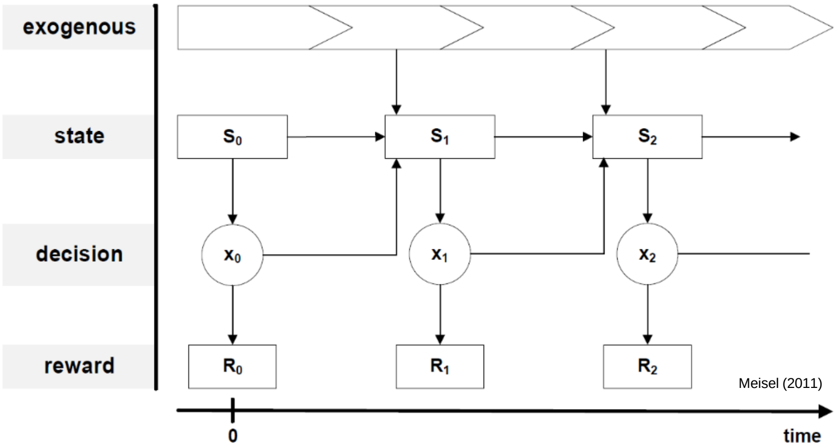


Agenda

- Dynamism
- Case study: Urban Delivery and Traffic Management
- **Preparation of Modelling**

Dynamic Decision Process

- Two processes:
  - Exogeneous: Changing information (stochasticity)
  - Endogenous: Adaption of plans (dynamism)



Meisel (2011)

→ A model should capture stochasticity and dynamism (over time)

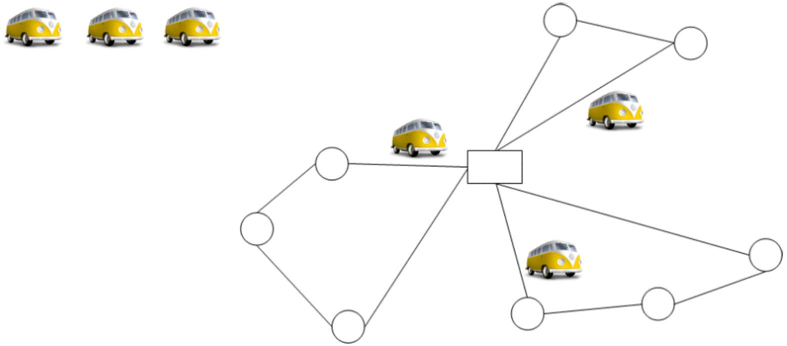
“Standard” Static Vehicle Routing Problem

- Depot, customers, vehicles
- Constraint: capacity
- Objective: minimize travel time

- Decisions
- Assignment of customers to vehicles
- Sequence of customers per vehicle

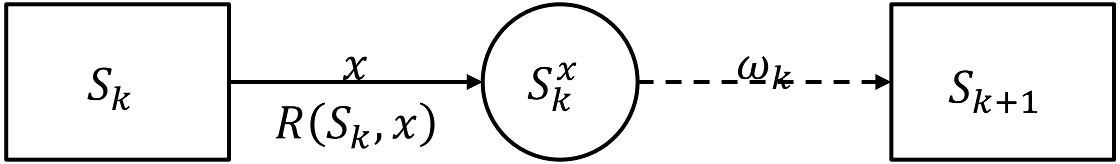
- Modeling: mixed-integer programming
- Solution: set of routes

- No consideration of temporal developments
- No consideration of stochasticity



$$
\begin{array}{lrl}
\text{minimize} & \displaystyle\sum_{(i,j)\in A}\sum_{k\in V} c_{ij} z_{ijk} & \\
\text{subject to} & \displaystyle\sum_{k\in V} y_{ik} \geq 1 & i \in C \\
& \displaystyle\sum_{(i,j)\in A} z_{ijk} = y_{ik} & i \in N,\ k \in V \\
& \displaystyle\sum_{(j,i)\in A} z_{jik} = y_{ik} & i \in N,\ k \in V \\
& \displaystyle\sum_{(j,i)\in A} x_{jik} - \sum_{(i,j)\in A} x_{ijk} = d_i y_{ik} & i \in C,\ k \in V \\
& x_{ijk} \leq L z_{ijk} & (i,j) \in A,\ k \in V \\
& y_{1k} = 1 & k \in V \\
& x_{ijk} \geq 0 & (i,j) \in A,\ k \in V \\
& y_{ik} \in \{0,1\} & i \in N,\ k \in V \\
& z_{ijk} \in \{0,1\} & (i,j) \in A,\ k \in V
\end{array}
$$

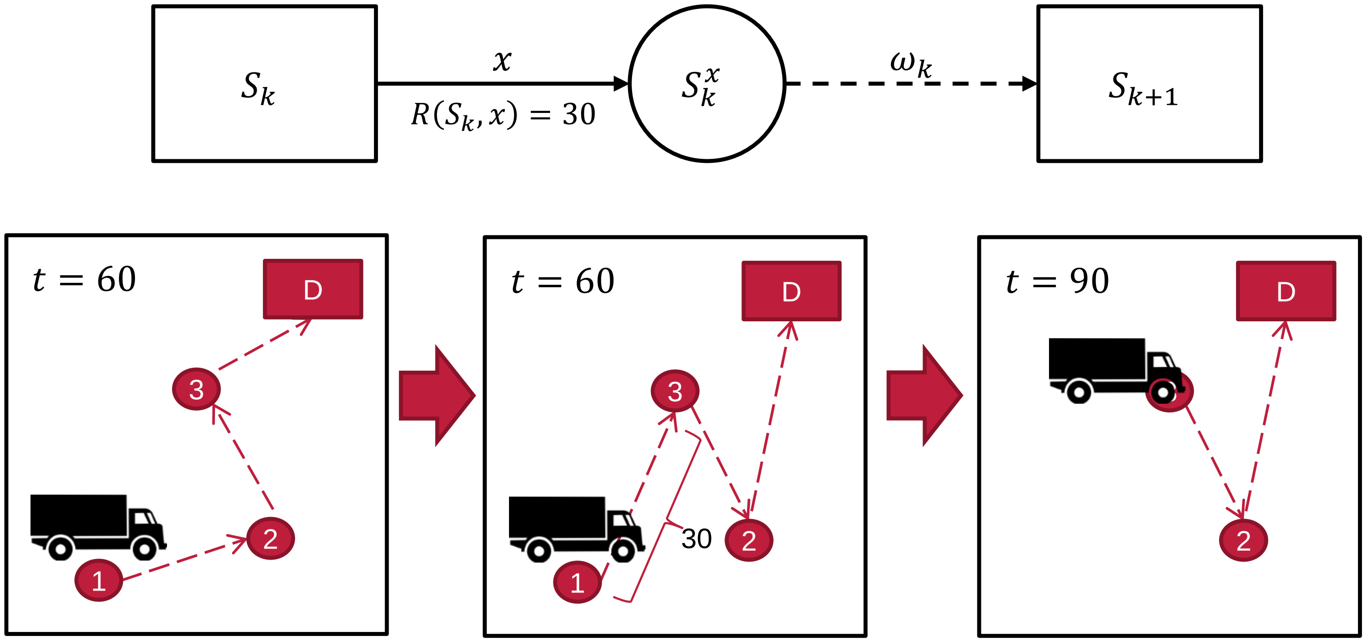
Markov Decision Process



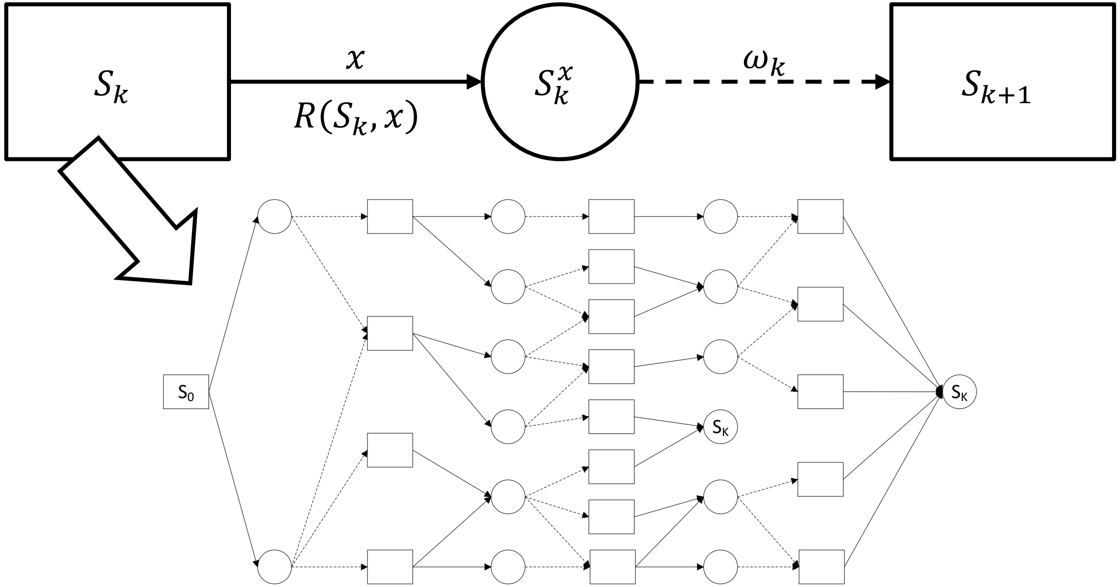
Components:

- Decision point k (either periodic or event triggered)
- Decision state $S_k \in \mathbf{S}$
- Decisions $x \in X$ (e.g., with Mixed-Integer Programming)
- Reward $R(S_k, x) \in \mathbb{R}$ (or costs as negative reward)
- Post-decision state $S_k^x \in \mathbf{S} \times X$
- Stochastic transition $\omega_k \in \mathbf{S} \times X \times \mathbf{S}$ (e.g., with probabilities $\mathbb{P}\colon \mathbf{S} \times X \times \mathbf{S} \rightarrow [\mathbf{0}, \mathbf{1}]$)

Example: Parcel Service



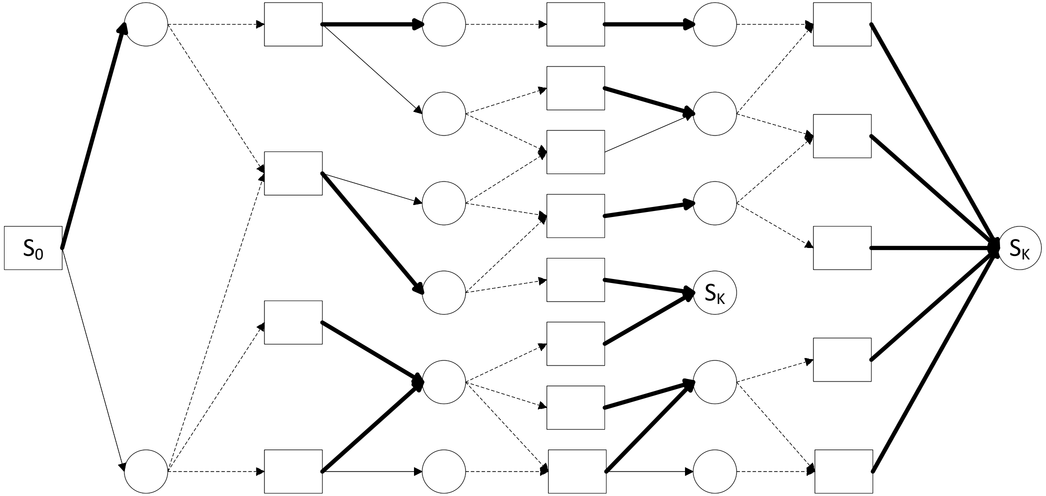
Problem Modeling: Markov Decision Process



- Initial state S_0
- Final state S_K

Solution: Decision Policy

- A decision policy assigns a decision to every state



- Objective: find optimal policy π^* maximizing expected overall rewards

Optimal Solution

- An optimal solution π^* satisfies:

$$\pi^* = \arg\max_{\pi\in\Pi} \mathbb{E}\left[\sum_{i=0}^{K} R(S_i, X_i^{\pi^*}(S_i))|S_0\right]$$

- Policy π^* represents the policy $\pi \in \Pi$, maximizing the expected reward
- Given initial state S_0
- $X_k^{\pi}(S_k)$ indicates the decision made by policy π in state S_k
- For minimization, we search for $\arg\min$

Summary

- Motivation und Definition: Dynamism

- Case Study: Dynamic re-planning reduces cost

- Modelling: Markov Decision Process

Outlook

- Anticipation: Integration of future developments in decision making

- Next lecture:
  - Markov Decision Process
  - Optimal solutions